

Lab 4 - Selenium Data Extraction Pipeline

Locator Strategy

For this lab activity, I chose the Fake Python Job Search website. Surprisingly, locating the element locators was quite straightforward. Every individual job card was held inside a “card-content” class which I used for my `items = driver.find_elements`. Within each content card I located the job title, company name, job location, and job posting date. The first three elements were categorized `By.CLASS_NAME` and the job posting date was categorized `By.TAG_NAME` since the date was contained in a time tag. I made sure to gather the text only and strip any whitespace.

When searching for element locators, I divide my screen with the website (fake jobs) and the HTML developer tool.

Resources Used

Since I wasn't able to get my Homework 6 program to work correctly, I never saw an output. Therefore, I relied heavily on YouTube videos, blog sites, and quick Google searches

- <https://stackoverflow.com/questions/55439502/iterating-through-multiple-elements-in-on-e-for-loop-using-selenium>
- <https://iproyal.com/blog/selenium-find-element/>
- <https://www.youtube.com/watch?v=1Bw2JHoOg5U>
- Google
 - “I'm getting this error, I believe it's about the items =
`driver.find_elements(By.CLASS_NAME, "card-content"` line: Item extraction failed:
Message: Compound class names are not allowed.; For documentation on this error,
please visit:
<https://www.selenium.dev/documentation/webdriver/troubleshooting/errors#invalidselectorexception>”

Challenges faced

Other than basic debugging issues (misspelling `executable_path` and forgetting parenthesis with `.strip`), I ran into two major issues:

1. At first, for `items = driver.find_elements` I used the class name for the entire section that held all the individual job cards. As a result, the program would open the Chrome browser, then quickly close it. To fix this error, I used the class name that held each individual job card, “card-content”. This way I could loop through the job elements within the individual job card.
2. My second error was about compound class names. I didn't realize that when I list out the element selectors the class names must be a single name, not the full unique class name. For example, I had to change the job title locator from `job_title =`

```
item.find_element(By.CLASS_NAME, "title is-5").text to job_title =
```

`item.find_element(By.CLASS_NAME, "title").text`. I made these changes to the rest of the locators, only using one word to find that particular class name element. After googling, I learned that Selenium's `By.CLASS_NAME` is expecting a single class name. It is also looking for one class that literally has a space in its name, which isn't valid for CSS. If I wanted to use the full unique class name, I would use the CSS Selector like:

```
item.find_element(By.CSS_SELECTOR, ".title.is-5").text
```

3. Interestingly, instead of using a class name for the `EC.presence_of_element_located` like I did in the previous homework, I used the ID name "ResultsContainer" which seemed to be a better choice.